

JEE MAIN REVISION MASTER: QUADRATIC EQUATION AND INEQUALITIES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 196 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Quadratic Equation And Inequalities - Core Application	196 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
-----------------	--

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Quadratic Equation And Inequalities - Core Application (Very High Yield — 196 PYQs)

Core Principle: Key problem pattern in Quadratic Equation And Inequalities.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $x \rightarrow 2$ $\lim_{x \rightarrow 2} \frac{\tan(x-2)(rx^2+(p-2)x-2p)}{(x-2)^2} = 5$ for some $r, p \in \mathbb{R}$. If the set of all possible values of p is $[a, b]$, then $a+b$ is..." [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "Let $\alpha, \alpha+2, \alpha \in \mathbb{Z}$, be the roots of the quadratic equation $x(x+2)+(x+1)(x+3)+(x+2)(x+4)+\dots+(x+n-1)(x+n+1)=4n$ for some..." [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "The number of the real solutions of the equation: $x|x+3|+|x-1|-2=0$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

Let $\lim_{x \rightarrow 2} \frac{\tan(x-2)(rx^2+(p-2)x-2p)}{(x-2)^2} = 5$ for some $r, p \in \mathbb{R}$. If the set of all possible values of q , such that the roots of the equation $rx^2 - px + q = 0$ lie in $(0, 2)$, be the interval $(\alpha, \beta]$, then $4(\alpha + \beta)$ equals :

- (A) 11
- (B) 13
- (C) 17 **[Correct]**
- (D) 21

Correct Answer: C

Solution Analysis: We are given $\lim_{x \rightarrow 2} \frac{\tan(x-2)(rx^2+(p-2)x-2p)}{(x-2)^2} = 5$. We have to first find the relation between r and p , and then determine all values of q for which the roots of $rx^2 - px + q = 0$ lie in $(0, 2)$. Step 1: Simplify the limit. Let $f(x) = rx^2 + (p-2)x - 2p$. Then the given limit becomes $\lim_{x \rightarrow 2} \frac{\tan(x-2)f(x)}{(x-2)^2} = 5$. Now, as $x \rightarrow 2$, $\tan(x-2) \sim x-2$. So the limit becomes $\lim_{x \rightarrow 2} \frac{(x-2)f(x)}{(x-2)^2} = \lim_{x \rightarrow 2} \frac{f(x)}{x-2} = 5$. For this limit to be finite,